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PROGRAMMABLE LOGIC CONTROLLER (PLC) PROGRAMMING

AN

INDIVIDUAL PROJECT

*Submitted for Partial fulfillment of the Requirement
For the award of the degree of*

BACHELOR OF SCIENCE

IN

INFORMATION TECHNOLOGY (ROBOTICS)

Submitted by

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Under the guidance of

Juliet Maikolo

Lecturer



DEPARTMENT OF INFORMATION TECHNOLOGY

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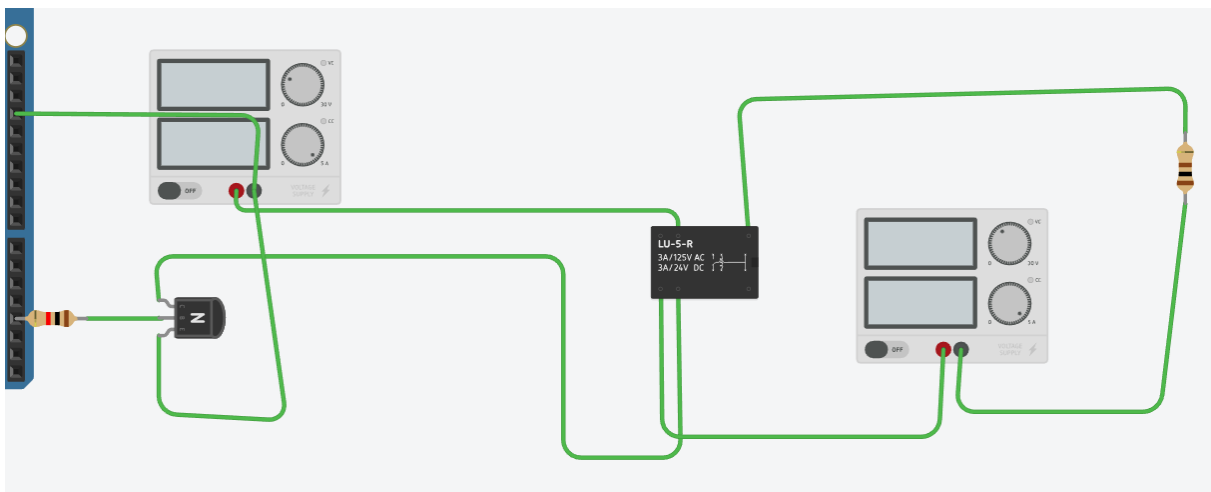
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Chapter 1: Question 1

1.1 Pulse-Width Modulation

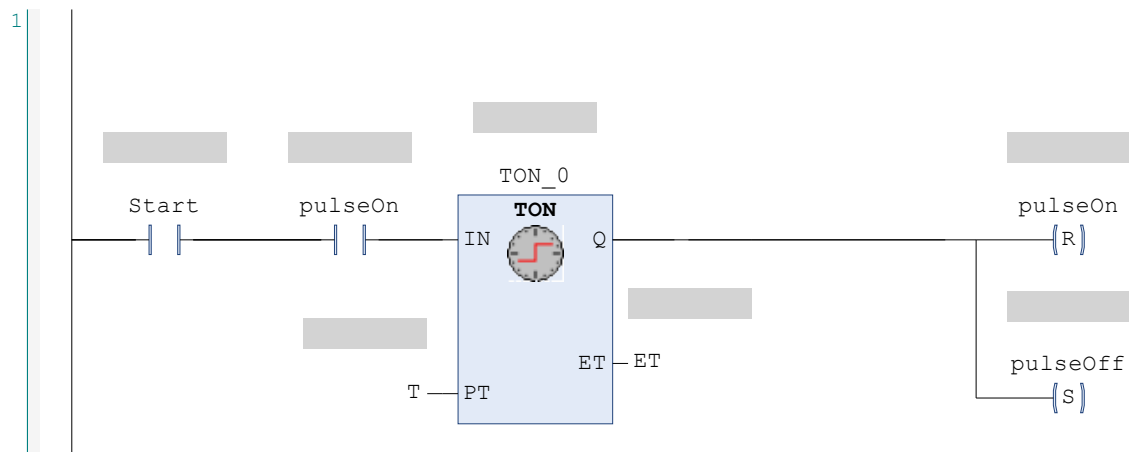
Pulse-Width Modulation (PWM) controls the average power delivered to a load by rapidly switching the supply off and on, with the average power over a set period being the current duty cycle. An example of PWM control can be seen below: Let a carrier signal be the sawtooth function oscillating between 0 and 1 with the frequency being the update frequency of a PWM signal (how often it can change between HIGH and LOW). an input reference is scaled down to between 0 and 1. And a the PWMs digital output signal can be gained from the comparison reference>carrier.

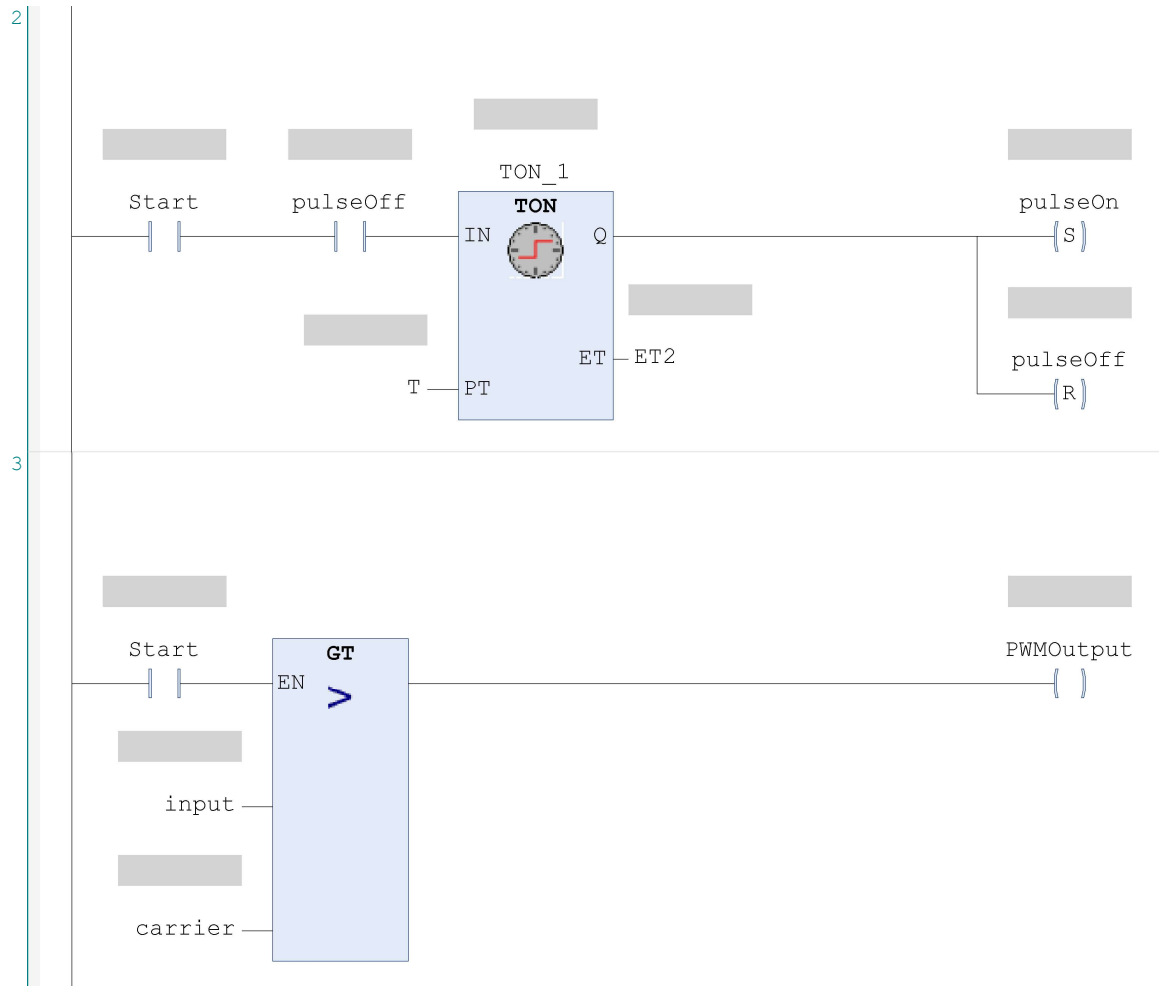
In the case of a heating element. The element itself is treated as a resistor. The resistor is powered by AC 220V and is switched on and off using a relay. For additional protection the relay can be energized by the power from a NPN transistor where the input is gained from the PLC through a current limiting resistor.

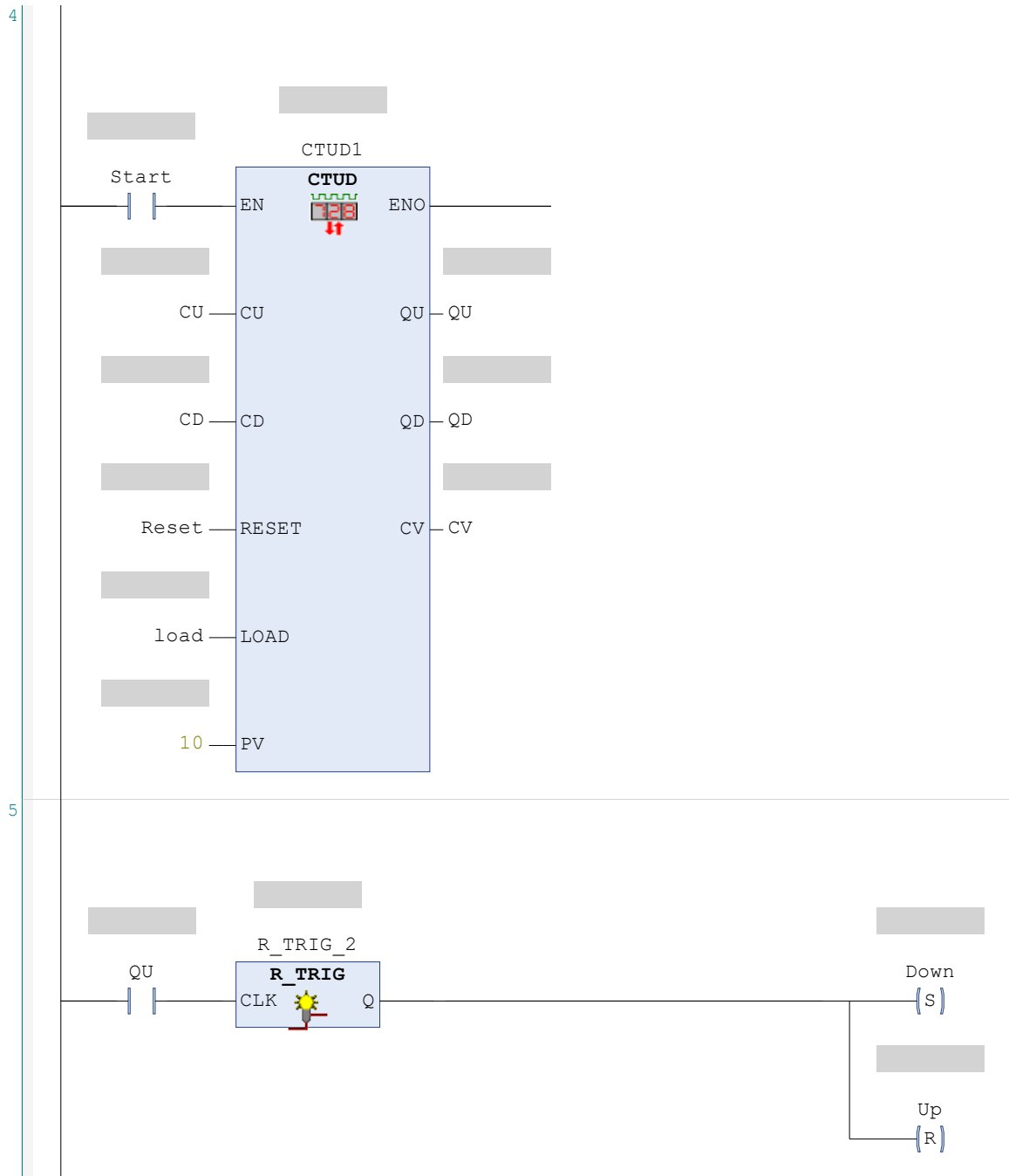


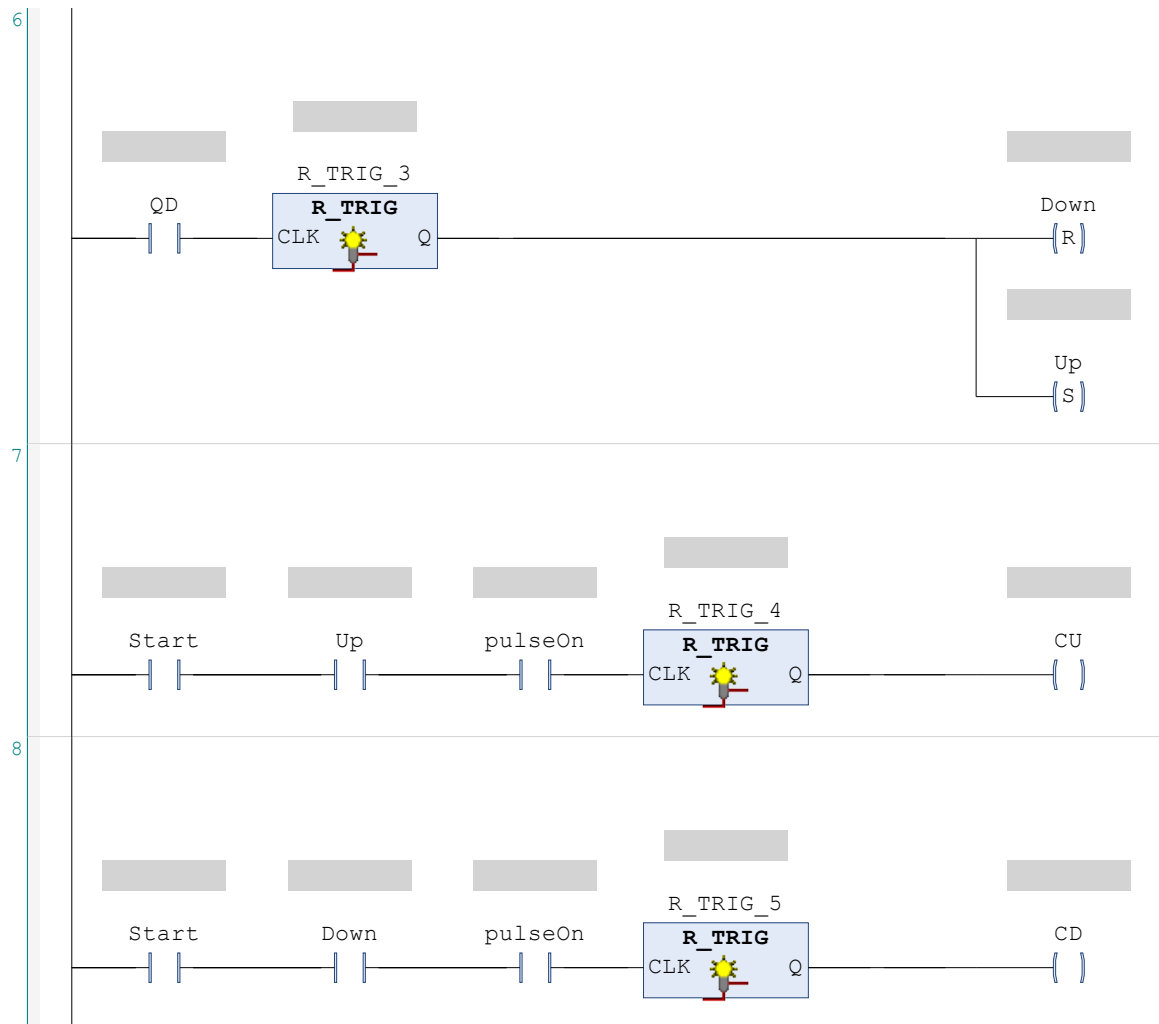
1.2 Ladder Logic Diagram

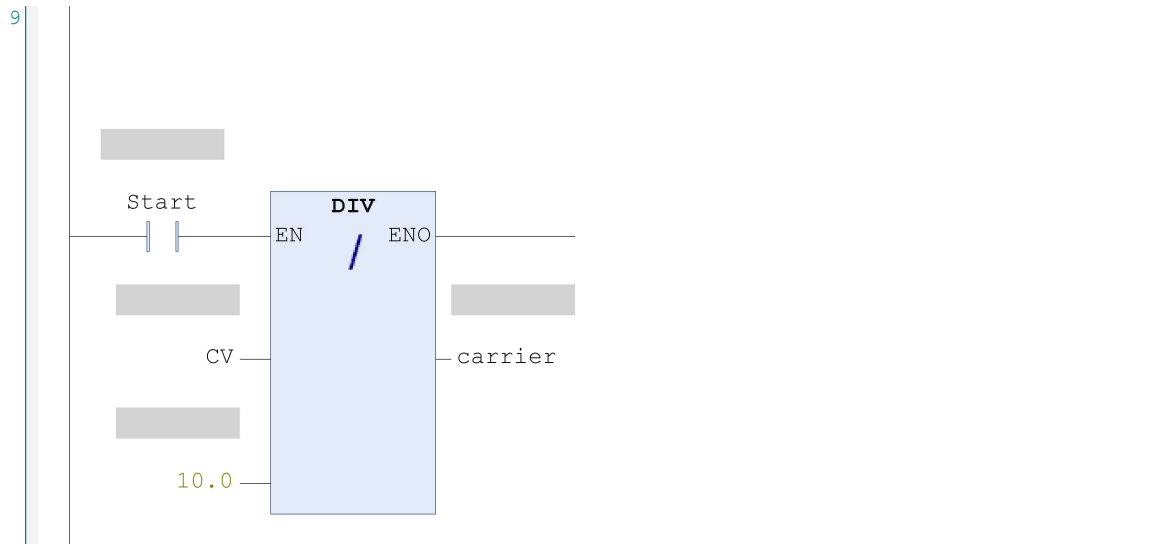
```
1  PROGRAM PLC_PRG
2  VAR
3      input : REAL ;
4      avgOutput : REAL ;
5      PWMOutput : BOOL ;
6      T : TIME := T#10MS ;
7      Start : BOOL ;
8
9      TON_0 : TON ;
10     R_TRIG_0 : R_TRIG ;
11     R_TRIG_1 : R_TRIG ;
12     ET : TIME ;
13     pulseOn : BOOL ;
14     pulseOff : BOOL ;
15     TON_1 : TON ;
16     ET2 : TIME ;
17     carrier : REAL ;
18     PWMTemp : BOOL ;
19     CU : BOOL ;
20     CD : BOOL ;
21     CV : WORD ;
22     CTUD1 : CTUD ;
23     QU : BOOL ;
24     QD : BOOL ;
25     R_TRIG_2 : R_TRIG ;
26     Down : BOOL ;
27     Up : BOOL ;
28     R_TRIG_3 : R_TRIG ;
29     R_TRIG_4 : R_TRIG ;
30     R_TRIG_5 : R_TRIG ;
31     Reset : BOOL ;
32     load : BOOL ;
33 END_VAR
34
```











1.3 Circuit Specifications

The cartridge heater used is a 300 W, 220 V resistive element with the following parameters:

- Rated Voltage: 220 V
- Power: 300 W
- Heater Size: 6 mm × 200 mm
- Lead Length: ±300 mm

The electrical resistance of the heater is calculated as:

$$R = \frac{V^2}{P} = \frac{(220 \text{ V})^2}{300 \text{ W}} \approx 161.3 \, \Omega$$

The full-load current is:

$$I = \frac{P}{V} = \frac{300 \text{ W}}{220 \text{ V}} \approx 1.36 \text{ A}$$

2SC3834 NPN:

- Collector-Emitter Voltage (Vce): 120V
- Collector-Base Voltage (Vcb): 200V
- Emitter-Base Voltage (Veb): 8V
- Collector Current (Ic max): 7A
- Collector Dissipation (Pc): 50W
- DC Current Gain (hFE): 70–220
- Transition Frequency (fT): 30MHz min
- Junction Temperature Range: -55°C to +150°C
- Operating Junction Temp (Tj): up to 135°C

Mini Relay 250V 16A - 5V Coil:

- Model: HF115F
- Coil Voltage: 5VDC
- Contact: Single contact (SPST)
- Mounting: PCB Through Hole

- Terminals: 5.0mm 1 pole 16A
- Rated Voltage (V): 250VAC
- Coil Power: Approx. 0.4
- Dielectric Strength (between coil contacts) (VAC 1min): 5000VAC 1min
- Operate Time: $\leq 15\text{ms}$
- Release Time: $\leq 8\text{ms}$
- Coil Resistance: $005:62\text{x}\square 1\pm 10$
- Max. Switching Voltage: 440VAC
- Max. Switching Voltage: 300VDC

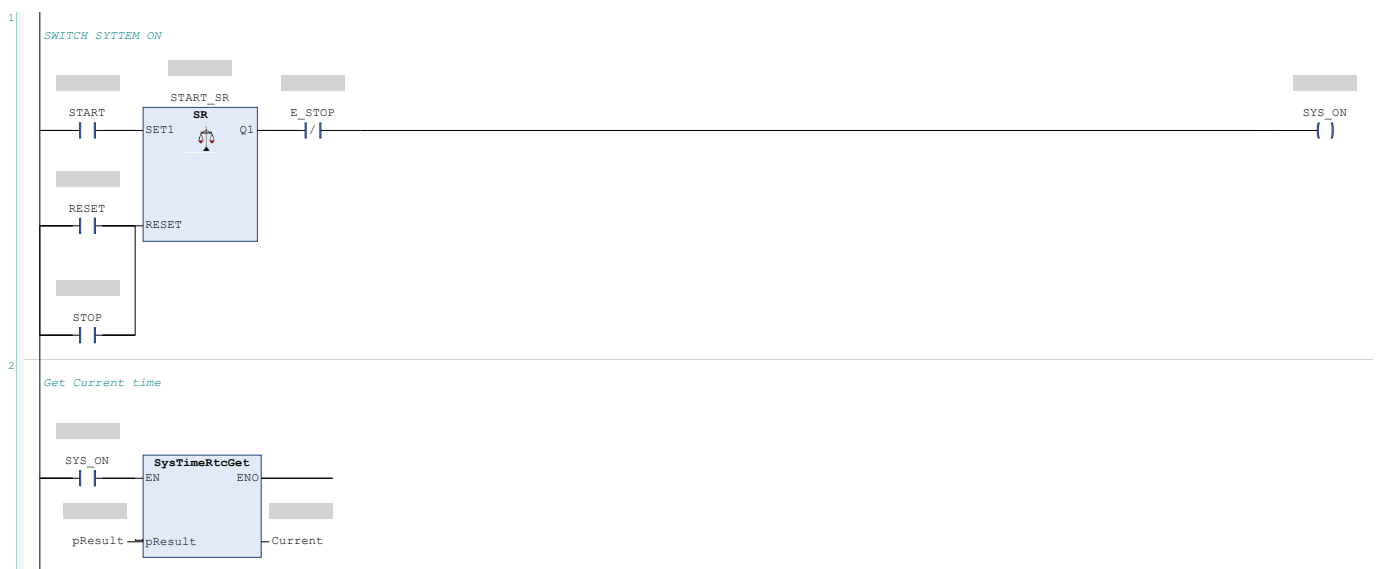
1.3.1 Switching Device Selection

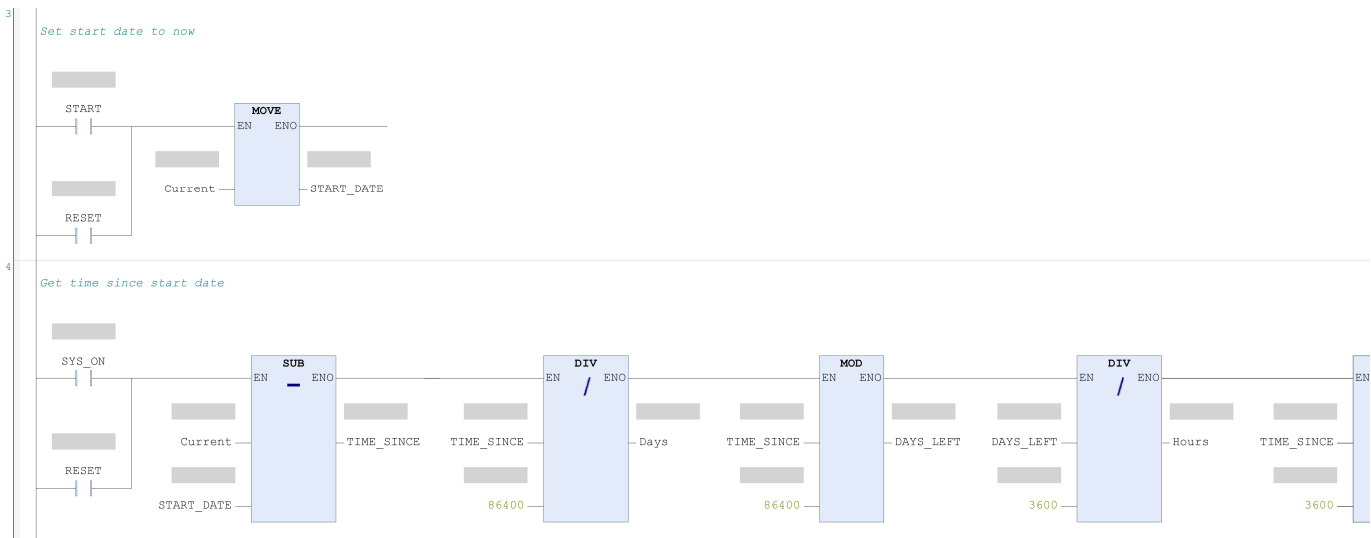
Chapter 2: Question 2

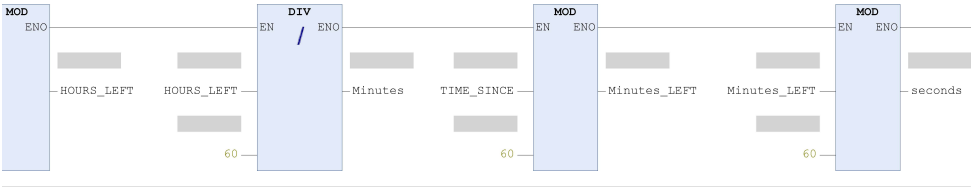
2.1 Ladder Logic diagram

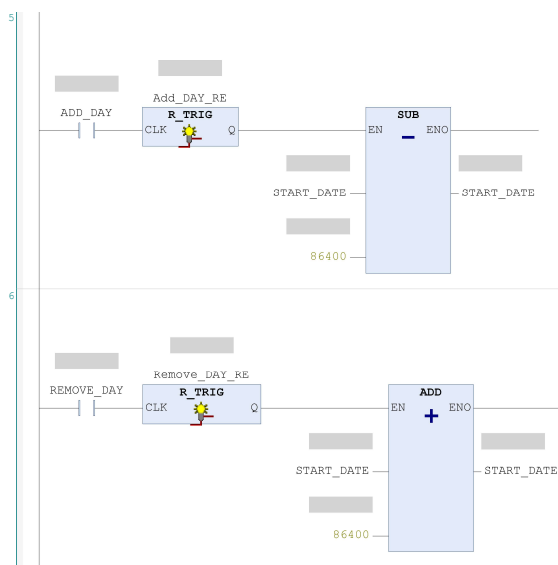
```
1  PROGRAM PLC_PRG
2  VAR
3      START_SR : SR ;
4      START : BOOL ;
5      RESET : BOOL ;
6      STOP : BOOL ;
7      E_STOP : BOOL ;
8      SYS_ON : BOOL ;
9
10
11      SET_START : BOOL ;
12      pResult : systimertc . RTS_IEC_RESULT ;
13      Current : DWORD ;
14
15
16      Days : DWORD ;
17      Hours : DWORD ;
18      Minutes : DWORD ;
19      seconds : DWORD ;
20
21      TIME_SINCE : DWORD ;
22      DAYS_LEFT : DWORD ;
23      HOURS_LEFT : UDINT ;
24      Minutes_LEFT : UDINT ;
25      START_DATE : DWORD ;
26
27
28      ADD_DAY : BOOL ;
29      Add_DAY_RE : R_TRIG ;
30      REMOVE_DAY : BOOL ;
31      Remove_DAY_RE : R_TRIG ;
32      ADD_HOUR : BOOL ;
33      REMOVE_HOUR : BOOL ;
34      Add_HOUR_RE : R_TRIG ;
35      Remove_HOUR_RE : R_TRIG ;
36      ADD_MINUTE : BOOL ;
37      Add_MINUTE_RE : R_TRIG ;
38      REMOVE_MINUTE : BOOL ;
39      Remove_MINUTE_RE : R_TRIG ;
40
41
42      FINISHED_RE : R_TRIG ;
43      Alarm : BOOL := TRUE ;
44      NoAlarm : BOOL ;
45      RESET_RE : R_TRIG ;
46      CYCLE_END : BOOL ;
47
48      TEMP : REAL := 35 ;
49      HUMIDITY : REAL := 53 ;
50
51      HEAT : BOOL ;
52      HUMIDIFIER : BOOL ;
53      FAN : BOOL ;
54
55
56      TEMP_LT_RE : R_TRIG ;
57      TEMP_LT_FE : F_TRIG ;
```

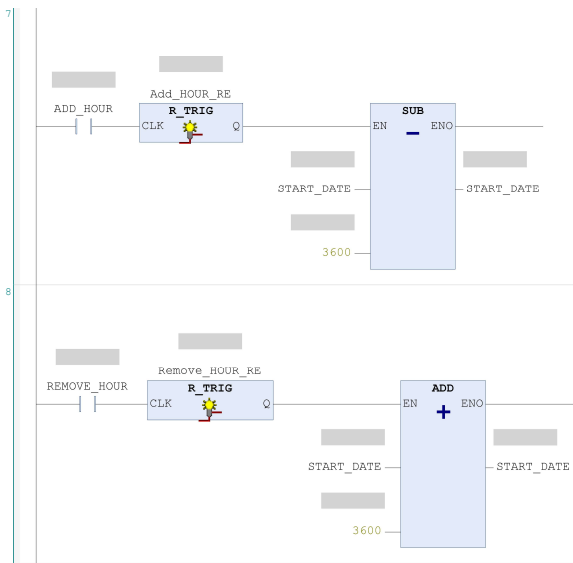
```
58
59     HEAT_ON : BOOL ;
60     HUMIDIFIER_ON : BOOL ;
61     FAN_ON : BOOL ;
62
63     HEAT_OFF : BOOL ;
64     HUMIDIFIER_OFF : BOOL ;
65     FAN_OFF : BOOL ;
66
67     TEMP_GT_RE : R_TRIG ;
68     TEMP_GT_FE : F_TRIG ;
69     HUMIDITY_GT_RE : R_TRIG ;
70     HUMIDITY_GT_FE : F_TRIG ;
71     HUMIDIFIER_ON_T : TON ;
72     HUMIDIFIER_ON_T_ET : TIME ;
73     HUMIDIFIER_OFF_RE : R_TRIG ;
74     HUMIDIFIER_OFF_T : TON ;
75     HUMIDIFIER_OFF_T_ET : TIME ;
76     HUMIDIFIER_ON_RE : R_TRIG ;
77     FAN_ON_T : TON ;
78     FAN_OFF_RE : R_TRIG ;
79     FAN_ON_T_ET : TIME ;
80     FAN_OFF_T : TON ;
81     FAN_ON_RE : R_TRIG ;
82     HEAT_ON_T : TON ;
83     HEAT_ON_T_ET : TIME ;
84     HEAT_OFF_T : TON ;
85     HEAT_OFF_T_ET : TIME ;
86     HEAT_OFF_RE : R_TRIG ;
87     HEAT_ON_RE : R_TRIG ;
88     FAN_OFF_T_ET : TIME ;
89     END_VAR
90
```

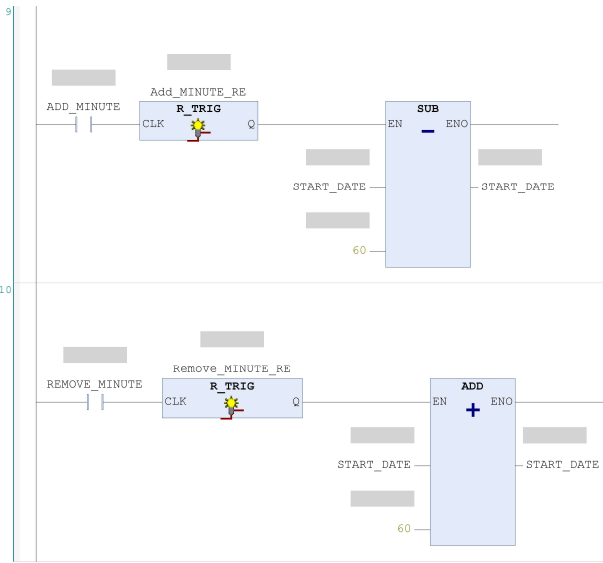


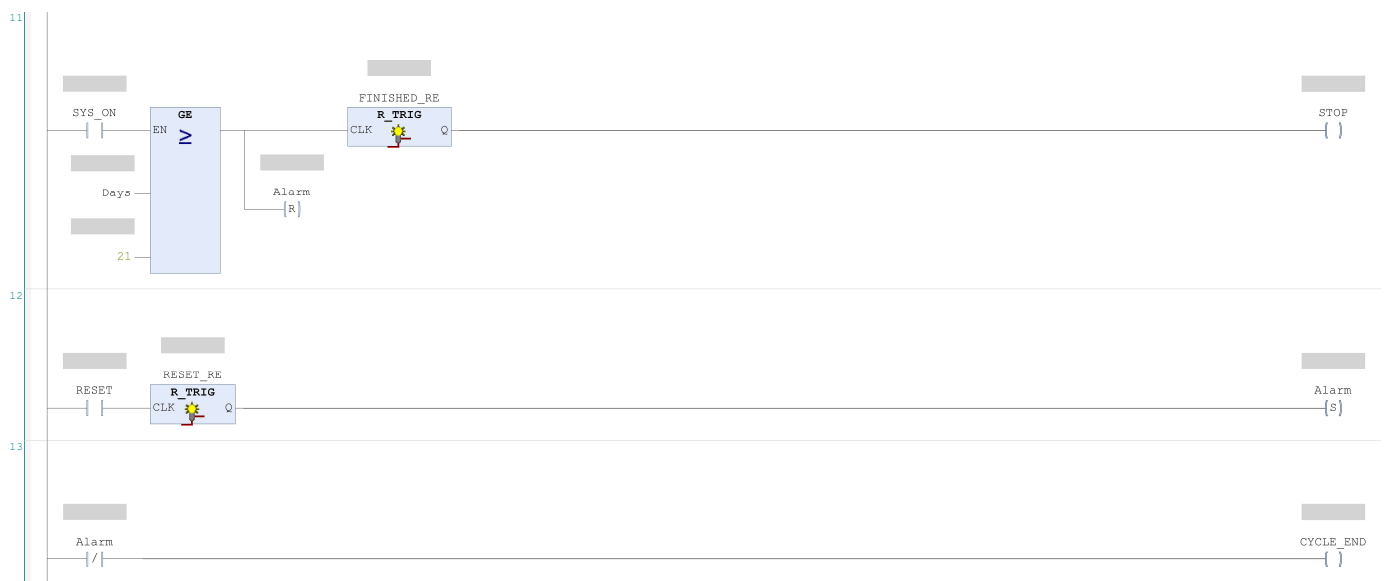


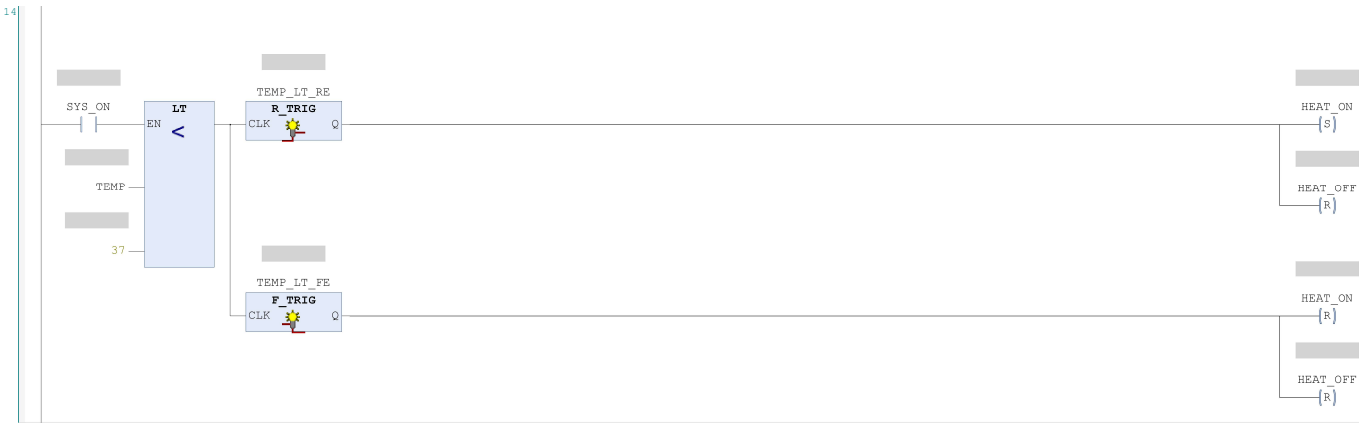


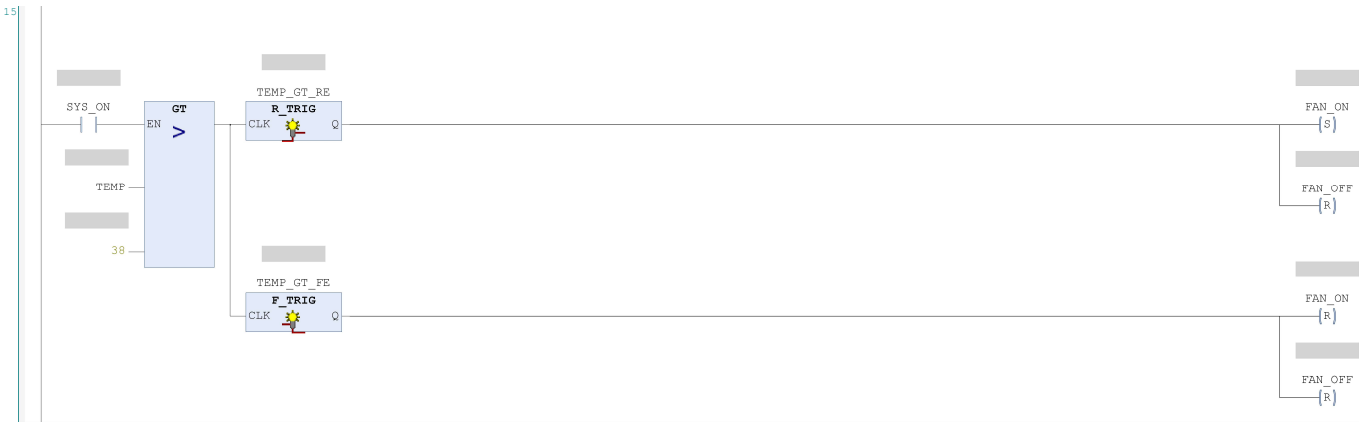


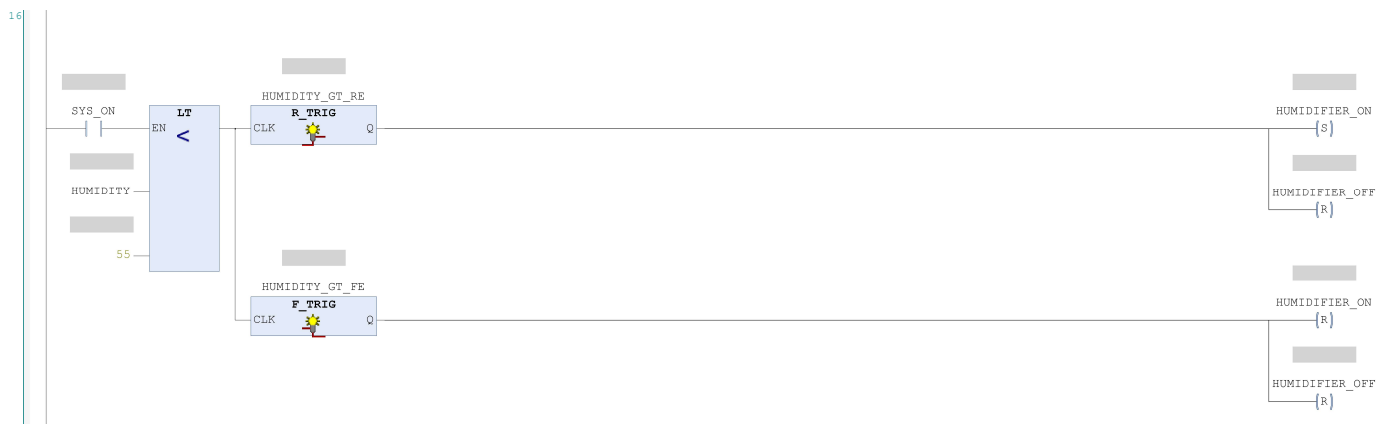


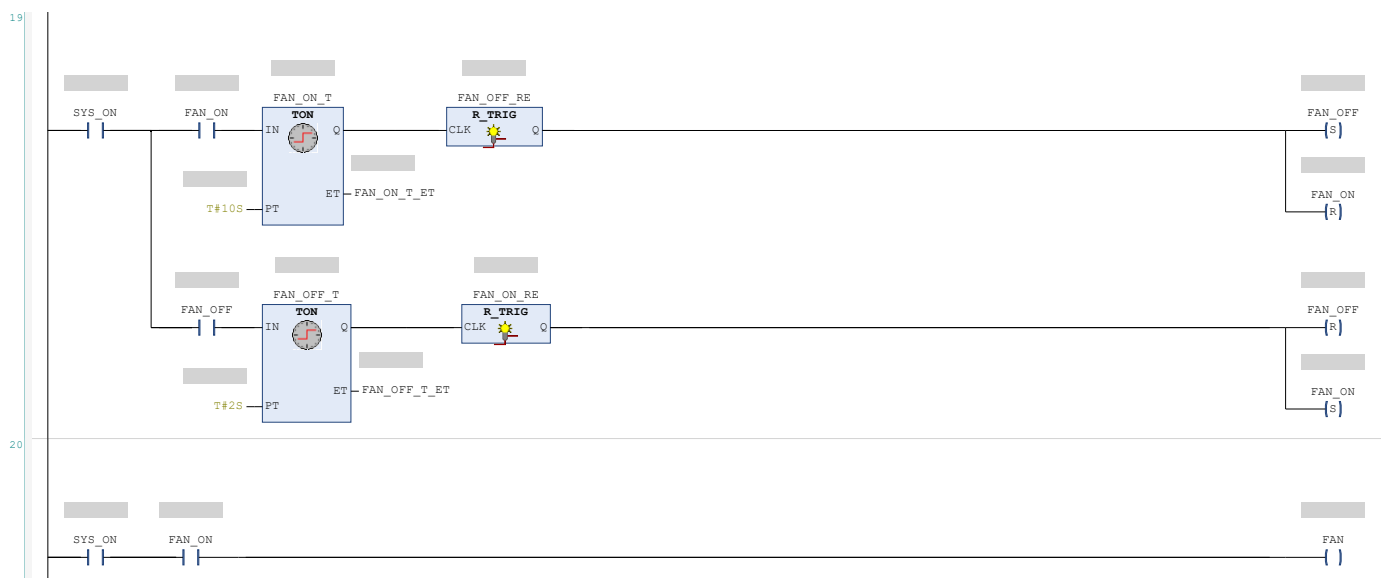


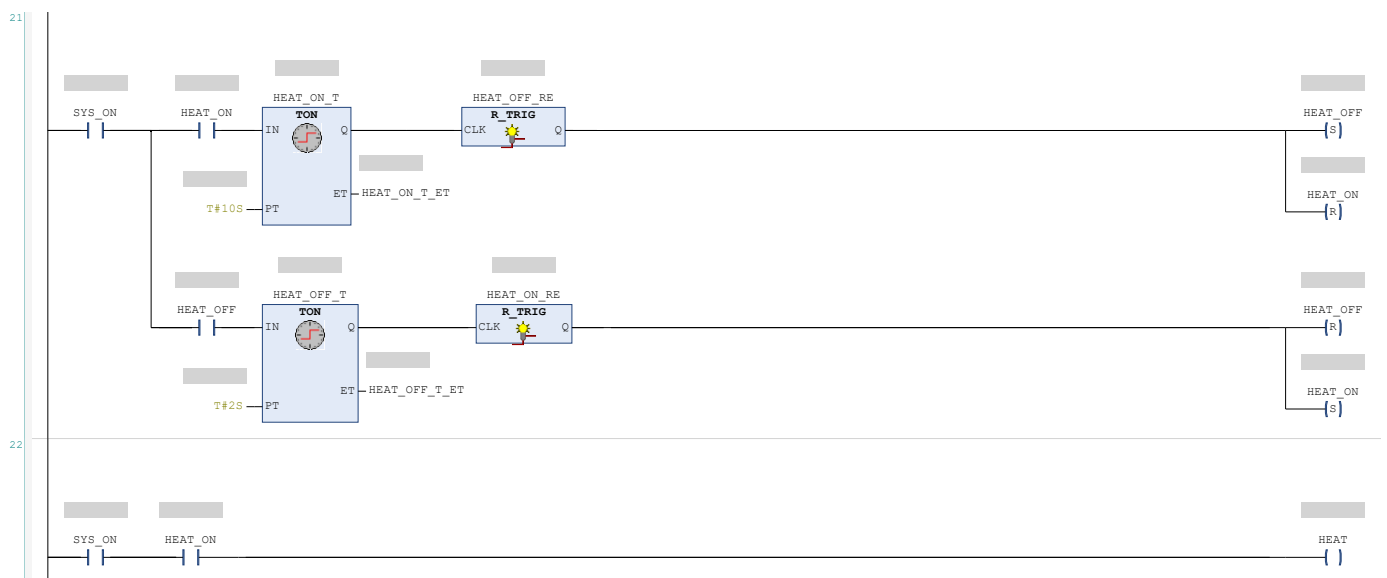












2.2 Explanation

1. A start/stop latch with a negated emergency stop button. The reset contact is set to the reset button and to stop contact that is in parallel to the reset contact is attached to the auto stop coil in rung 11
2. when the system is powered, the sysTimeRTCGet function block gets the current time on the device and places it in the current variable. Current is the time in seconds since 1970-01-01 00-00:00
3. if either start or reset is pressed then the time in current is saved to start_date
4. if the system is running or while reset is held down, the following calculations are done:
 - $TIME_SINCE = Current - START_DATE$
 - $Days = \frac{TIME_SINCE}{86400}$
 - $DAYS_LEFT = TIME_SINCE \% 86400$
 - $HOURS = \frac{DAYS_LEFT}{3600}$
 - $HOURS_LEFT = DAYS_LEFT \% 3600$
 - $MINUTES = \frac{HOURS_LEFT}{60}$
 - $SECONDS = TIME_SINCE \% 60$
5. manually add day to time counter by subtracting a days worth of seconds from START_DATE
6. manually removing day to time counter by adding a days worth of seconds from START_DATE
7. manually add hour to time counter by subtracting a hour worth of seconds from START_DATE
8. manually removing hour to time counter by adding a hour worth of seconds from START_DATE
9. manually add Minute to time counter by subtracting a Minute worth of seconds from START_DATE
10. manually removing Minute to time counter by adding a Minute worth of seconds from START_DATE
11. while system is on check if days are 21 or more, if so the alarm goes off and the system is stopped without being reset

12. when reset is pressed the alarm is rearmed
13. when the alarm is triggered, the end cycle led is on
14. if the temperature sensed goes below 37degC or switches on, then the clock cycle for the heating element is started. If the temperature sensed goes above 37degC the clock cycle is switched off.
15. similarly if the temperature sensed goes above 38degC or switches on while above 38degC, then the clock cycle for the fan is started. If the temperature sensed goes below 38degC the clock cycle is switched off.
16. similarly if the humidity sensed goes below 55% or switches on, then the clock cycle for the humidifier is started. If the humidity sensed goes above 55% the clock cycle is switched off.
17. this is the clock cycle for the humidifier
18. if the humidifier on pulse is high then the humidifier is on
19. this is the clock cycle for the fan
20. if the fan on pulse is high then the fan is on
21. this is the clock cycle for the heating element
22. if the heating element on pulse is high then the heating element is on

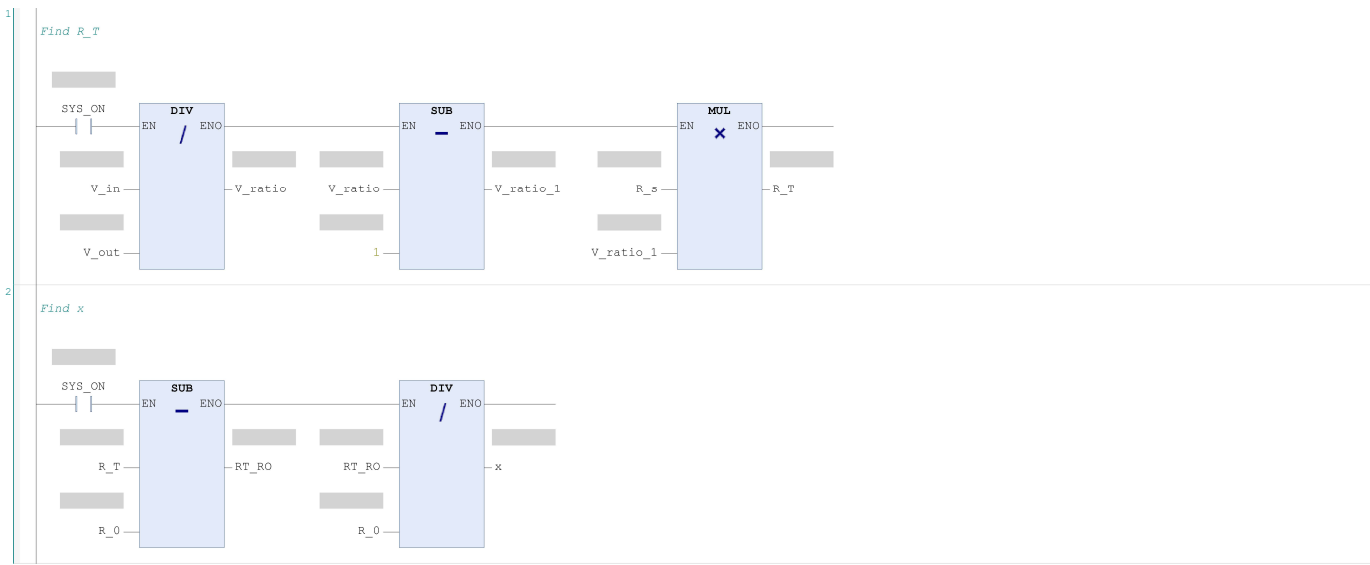
Chapter 3: Question 3

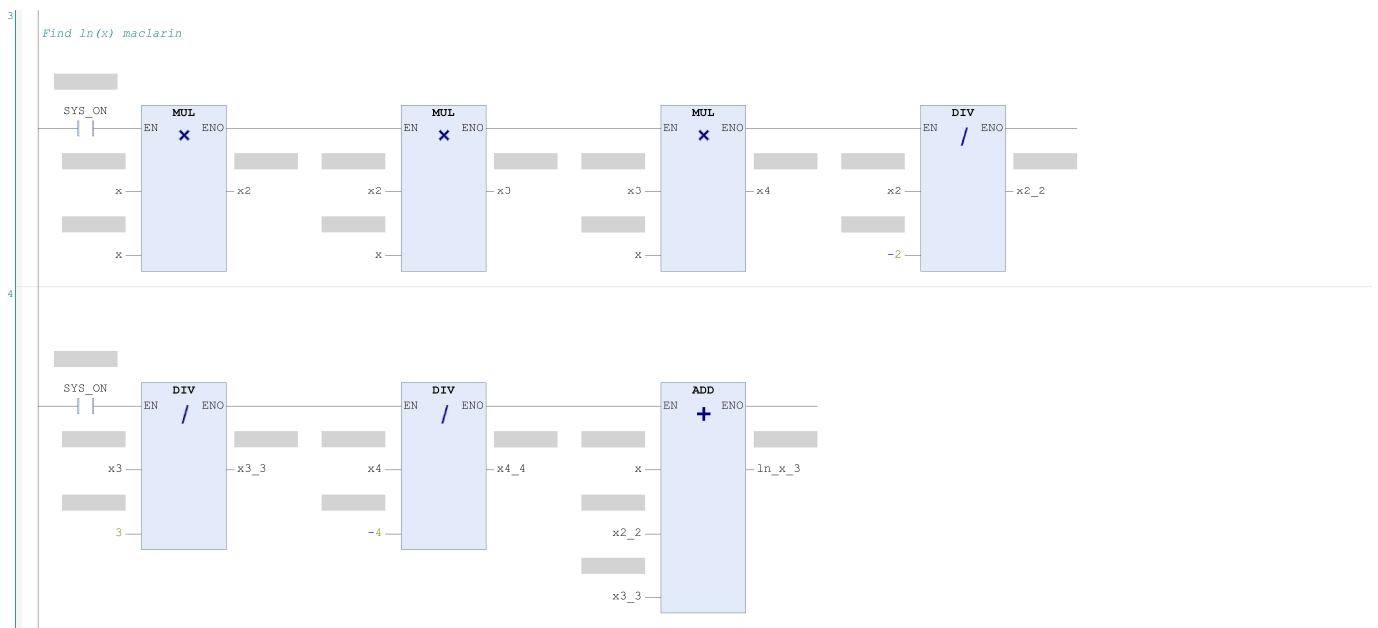
3.1 Visualization

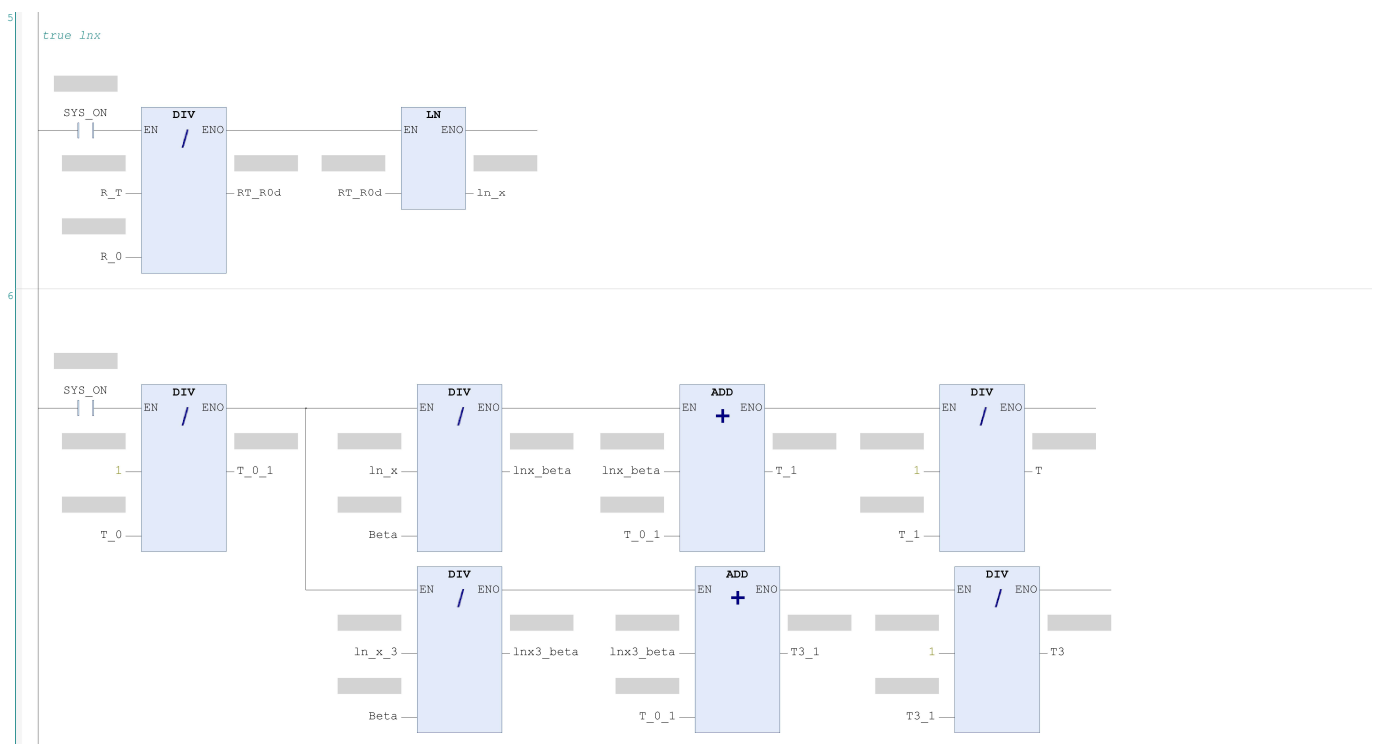


Chapter 4: Question 4

```
1  PROGRAM PLC_PRG
2  VAR
3      SR_0 : SR ;
4      START : BOOL ;
5      STOP : BOOL ;
6      SYS_ON : BOOL ;
7
8      VOLTAGE : REAL ;
9
10     T_0 : REAL := 298.15 ;
11     R_0 : REAL := 10000 ;
12     Beta : REAL := 3950 ;
13     R_T : REAL ;
14     T_c : REAL ;
15     V_IN : REAL := 5 ;
16     R_s : REAL := 10000 ;
17     V_out : REAL := 3 ;
18
19     V_ratio : REAL ;
20     V_ratio_1 : REAL ;
21     RT_RO : REAL ;
22     x : REAL ;
23     x2 : REAL ;
24     x3 : REAL ;
25     x2_2 : REAL ;
26     x3_3 : REAL ;
27     ln_x : REAL ;
28     x4 : REAL ;
29     x4_4 : REAL ;
30     ln_x_3 : REAL ;
31     T_0_1 : REAL ;
32     lnxBeta : REAL ;
33     lnX3_beta : REAL ;
34     T_1 : REAL ;
35     T3_1 : REAL ;
36     T : REAL ;
37     T3 : REAL ;
38     T_c3 : REAL ;
39     RT_ROd : REAL ;
40 END_VAR
41
```









$$R_T = R_s(\frac{V_{in}}{V_{out}} - 1)$$

$$x = \frac{R_T - R_0}{R_0}$$

$$\ln(x) = \frac{R_T}{R_0} = \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3}$$

$$T = \frac{1}{\frac{1}{T_0} + \frac{\ln(\frac{R_T}{R_0})}{\beta}}$$

$$T_C = T - 273.15$$

Bibliography